Replies to reviews of the paper *blocking: An R Package for Blocking of Records for Record Linkage and Deduplication*

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1 General comments

We have made the following changes to the paper, and detailed replies are provided in the subsequent sections. We can summarise it as follows:

- Clarified the description of entity resolution and blocking, including the role of unique identifiers, the distinction between blocking and final linkage, and an explanation of ANN algorithms and shingles.
- Revised the evaluation metrics section: added lay descriptions and blocking-specific interpretations for all metrics, corrected the reduction ratio formula, added a reference for the reduction ratio, and noted the critique of the F1 score by Christen and Hand.
- Corrected errors in the manuscript, including the number of comparisons (from 6 billion to 700 billion), the FNR values when comparing default and increased epsilon, and minor typographical issues throughout.
- Extended the record linkage case study (Section 3.1) to include a complete entity resolution workflow using the Expectation-Maximization algorithm by Fellegi and Sunter via the `reclin2` package.
- Added comments to code listings and moved lengthy code blocks (Table 2 generation, confusion matrices, shingle and embedding comparisons) to Supplementary materials.
- Added a discussion of Figure 1 (density plot), explaining the bimodal distribution of distances for true matches and its implications for blocking.
- Clarified the requirements for the `true_blocks` argument in both the manuscript and the package documentation.

- Updated the summary to include a paragraph on Privacy-Preserving Record Linkage as a direction for future work, and a note on potential scalability via DuckDB.
- Updated references: replaced the Soundex citation with Knuth (1973), added Christen (2012) references for early blocking work, and added Steorts et al. (2014) for LSH and shingles.
- Broadened the framing of the package to emphasise that it is not limited to the `reclin2` package or to the foreign-national population context.
- Supplementary materials include annotated R scripts for reproducing the analyses presented in the paper:
  - `1-embeddings-example.R` – Blocking for record linkage using the `blocking` package on the `foreigners` dataset, comparing text-based blocking (with 2- and 3-character shingles) and embedding-based blocking via a local Ollama model.
  - `2-codes-for-table-2.R` – Comparison of approximate nearest neighbour algorithms (nnd, hnsw, annoy, lsh, kd) from the `blocking` package and k-means LSH from the `klsh` package for deduplication on the `RLdata500` dataset (Table 2).

2 Reviewer 1

- Amendment: "...is essential for estimation [of data] based on multiple sources...". Feedback: Its importance is more about bringing together the data needed for the estimation (or other analyses) not the estimation itself. Currently reads like the latter.

  We have followed this recommendation and updated the text.

- Amendment: "foreign-nationals" not "foreign-born". Feedback: Nationals can be foreign-born and will have the national IDs.

  We have revised the manuscript accordingly.

- Amendment: "... link records without [unique/personal] identifiers ...". Feedback: Strictly speaking, an identifier is always needed for record linkage. Sex, Age, Address etc are al identifier albeit less personal or unique.

  We have followed this recommendation and updated the text.

- Amendment: "To [validate the blocking strategy], a subset ...".

  We have revised the manuscript accordingly.

- Feedback: The phonetic argument is actually used for blocking in Recordlinkage. See except: *Note that the phonetic function, if requested, is applied before the blocking process, therefore the equality restrictions imposed by blocking apply to phonetic codes Sariyar & Borg 2010.*

  We have removed the sentence as suggested.

- Feedback: Clarify if B(i) in RR(dedup) is count of unique records as with B(ix) and B(iy) in RR(reclin).

  As also suggested by Reviewer 2, we have removed the word "unique" to avoid confusion with unique individuals.

- Feedback: I suggest clarifying that "Recall" is "Sensitivity".

  We have followed this recommendation and added the term "Sensitivity" in parentheses next to "Recall".

- Feedback: Include a lay description of what information each metric provides. Similar to what's you have for TP, FP, TN & FN.

  We have provided lay descriptions for all evaluation metrics, along with their specific interpretation in the context of blocking.

- Feedback: I followed your example but my `result_reclin` object has 6469 blocks and a slightly different distribution of block sizes. Please include why this could be the case given that a user has followed the same example above and used the default seed. Maybe a different versions in one of its dependent packages?

We have tested the code on multiple environments and consistently reproduced the reported results. We agree with the Reviewer's assessment that the discrepancy is likely caused by different versions of dependent packages.

The default method used in our package relies on the `rnndescent` package which imports the `dqrng` package. Probably that was the issue, we suggest to update the `rnndescent` package to the recent version (0.1.8) because there were some issues with the `dqrng` package.

- Feedback: Make clear the requirement for the `true_block` matrix in the paper and function's help file. One I noticed is that the colnames must be "x", "y" and "block"

  We have revised both the manuscript and the documentation accordingly.

- Feedback: "That decreases the FNR to 3.25%." is logically incorrect. Maybe a typo in your value? Running the sample code gives me a reduction in FNR from 3.2218 to 3.1318.

  We apologize for the error which was due to copying the code from previous example. We have corrected the manuscript.

- In practice, there are likely to be low-risk simple blocking attributes acceptable to users so they can prevent maxing out computing resources when working with larger datasets. Simple blocking on these attributes before the ANN search will be very helpful (and possibly critical) for this. I note the package authors have an `on_blocking` argument to achieve this but it's not yet supported as of v1.0.1. In the mean time, I recommend the inclusion of a supported/official way to append (rbind) either 'blocking' objects or their "result" data.frames. The benefit of this is that it will allow users to perform simple blocking by themselves through separate calls of blocking() per their chosen blocking attribute. These results, in particular their blocks, can then be appended together into one "blocking" object. Users can then proceed with the rest of the work flow as described in the paper. Sample code to demonstrate this below. Basic benchmarking for the example below indicates a 50% reduction in peak RAM usage which is considerable when working with Big Data.

```
# Example of simple blocking by country before ANN search
blocking_lst <- list()
countries <- unique(foreigners$country)
# Loop through separate blocking() calls per country
for(x in countries){
  # Restrict (i.e. block) to the particular country
  x_index <- which(foreigners_1$country == x)
  y_index <- which(foreigners_2$country == x)

  tmp_x <- foreigners_1$txt[x_index]
  tmp_y <- foreigners_2$txt[y_index]
  # ANN search
  tmp_reclin <- blocking(x = tmp_x, y = tmp_y)
  # Update x,y to original index
  tmp_reclin$result[, ‘:=‘(x = x_index[x], y = y_index[y])]
  blocking_lst[[x]] <- tmp_reclin
  }
# Unofficial appending of blocks.
```

```
pseudo_recline <- blocking_lst[[1]]
pseudo_recline$result <-
  do.call(rbind, lapply(blocking_lst, function(x) x$result))
# An official and supported method is instead recommended
to allow continued integration with other packages
```

We appreciate the comment and the provided benchmark. We acknowledge the benefits of this approach for large datasets and will consider implementing such an option in a future version of the package. Currently, as noted by the Reviewer, users can apply this blocking strategy manually by splitting the data.

Please note that country of citizenship may differ between data sources. In our practice (at statistical office) we have encountered situations where the same person had XXX citizenship in the population register but in social insurance register had ZZZ citizenship (provided by employer). These situations are not common but we would like to leave the decision to researcher.

3 Reviewer 2

- In the abstract: Entity resolution "aims to link records without common identifiers" is not well formulated. Entity Resolution aims to link records that refer to the same entity in absence of unique identifiers. Entity Resolution uses variables, common to all records, which allow to identify entities approximately, while blocking usually uses a subset of them in order to reduce the complexity of the task. Same comments for the beginning of the introduction, be clearer on the objective of entity resolution and on the role of blocking in the entity resolution pipeline.

  We have updated the text to explicitly mention the absence of unique identifiers, as suggested. Following this clarification, we reviewed the description of blocking and believe its role in the pipline is now sufficiently clear.

- Blocking is useful in a lot of applications of record linkage or deduplication; it should be clear that the running data example on 'foreigners' is an illustration and that the package may be useful beyond this particular application.

  We have revised the text to explicitly state that the foreign-national populations context is intended solely as an example.

- The number of pairs of records to compare in entity resolution given n records is $\frac{n(n-1)}{2}$ so that $\frac{1.206.000\times(1.206.000-1)}{2} \approx 700.000.000.000 = 7 \times 10^{11}$ is rather 700 billion more that 6 billion. Is there a reason to choose 6 billion as a reference?

  Thank you for spotting this! We apologize for the error. We have corrected the manuscript.

- The idea of blocking on nearest neighbours does not only find motivations in embeddings and LLM, there is some earlier work from Peter Christen on the topic, where more blocking techniques are proposed.

  We have added two references to the early work of Peter Christen:

  * Christen, P. (2012). A survey of indexing techniques for scalable record linkage and deduplication. IEEE Transactions on Knowledge and Data Engineering, 24(9), 1537–1555. https://doi.org/10.1109/TKDE.2011.127
  * Christen, P. (2012). Data matching: Concepts and techniques for record linkage, entity resolution, and duplicate detection. Springer Science & Business Media. https://doi.org/10.1007/978-3-642-31164-2

- Blocking is used to reduce the complexity of entity resolution, and it can also be used as a way to assess the difficulty of the record linkage or deduplication task. A naive way of performing record linkage for example could be to link together all the pairs for which all linking variables match (blocking on all linking variables). By doing so, one can obtain an upper bound on the number of false positives a record linkage algorithm would identify (as the false positives identified would correspond to pairs of records with matching information). This might be another argument to motivate the use of blocking methods in the explorative phase of the entity resolution task.

  Thank you for this comment.

- The authors mention missing information and the package seems to handle them, could the authors develop on how are missing values taken into account? (I come back to this topic in a comment on section 3.2).

  Missing values are treated as empty strings, as visible in the example in Section 3.1.

- The reduction ratio evaluates the reduction in complexity thanks to blocking and does not require to know the number of 'unique records' often unknown in real data problems. Unique records may refer to the underlying unique individuals whose records may have been duplicated. The authors may rephrase to avoid any confusion.

  We have rephrased the text by removing the word "unique".

- Blocking should not be used to provide the final linkage as the authors mention. In particular, blocking would ideally eliminate pairs which we do not want to discover (FP and TN) while including the pairs which we want to discover (TP and FN), though we cannot expect blocking to discriminate between the positives (FP and TP) or between the negatives (FN and TN).

  - Recall is expected to be high as blocking should discover all pairs one wants to compare in entity resolution. Also, 'recall' is more common in the machine learning field and is often called 'sensitivity' in the statistical literature, though in blocking it seems to be more common to talk about 'pairs completeness'.
  - Precision (or 'positive predictive value') is expected to be low as blocking is not meant to provide the final linkage and therefore does not distinguish TP from FP. It seems to be more common to talk about 'pairs quality' in the blocking literature.
  - F1-score might not be relevant in that context (since high recall and low precision are expected).
  - Accuracy may be relevant as it evaluates the ability to discriminate true (positives and negatives) from false (positives and negatives).
  - Specificity is expected to be high as the obvious non-links (pairs of distinct records) should be spotted as TN and, a lot of TN are expected from the set of all pairs to be compared.
  - FNR = 1 - recall, choose one or the other.
  - FPR = 1 - specificity, choose one or the other.

  In these regards, not all the metrics are relevant for blocking and they should be used with caution. It might be good as well to simply show the confusion matrix (i.e. the amount of TP, FP, TN, FN), or the partial confusion matrix without TN (lots of TN are expected from the set of all pairs).

  We have provided descriptions for all evaluation metrics, along with their specific interpretation in the context of blocking. We have focused on FPR and FNR instead of recall and specificity. However, in the manuscript, we have mentioned both recall and specificity and highlighted their relation to FPR and FNR. We have decided to keep all metrics in the package to allow users to follow their preferences.

  Regarding the confusion matrices, we have added the code generating Table 2 to the Supplementary materials. It now also outputs confusion matrices for all compared methods (see 2-codes-for-table-2.R code in Supplement).

- The SPLink documentation (method developed in python) proposes ways to analyse and visualise the effect of blocking. It would be very useful for users to see such evaluation graphics in the paper (also relevant for the python version of this article/package). Moreover, it is good to note that the number of unique values of the linking variables have specific influence on the overall number of pairwise comparisons, so that blocking should be used hierarchically and maybe not on all variables.

  We appreciate the reference to the visualizations provided by the SPLink package. However, since the `blocking` package operates on the blocking variables concatenated into a single string, generating such variable-specific plots is technically difficult in our framework. Regarding the second point, we agree that variable selection and hierarchical strategies are important; our package supports this by allowing users to flexibly choose which variables to include, leaving the methodological decisions to the user.

- In the blocking results output 'Number of columns used for blocking: 1232' I interpret that 1232 shingles are created to encode the information contained in the records before searching for nearest neighbours. Can the authors elaborate on that? Same in the vignette of the CRAN package, 'it was based on 429 columns (2 character shingles)' is unclear.

  The Reviewer's interpretation is correct. This number represents the number of columns in the matrices used to represent the data, which can also be provided explicitly by the user as `x` and `y`. Since these matrices can also pertain to embeddings, we have decided not to explicitly mention shingles in the `blocking` function output to maintain generality. For an example of embeddings via the `ragnar` package see Sumplementary Materials: 1-embeddings-example.R.

- When comparing 'res_reclin' and 'res_reclin2' with the default epsilon (0.1) and the increased epsilon (0.5) in order to be more stringent on the blocking output (?), I suspect that the authors want to illustrate how the FNR can be improved ('That decreases the FNR to 3.25%'). However the two outputs are identical: the FNR with default epsilon is 3.25%, as is the one obtained with epsilon set to 0.5, this is probably a mistake. The vignettes provide examples where the FPR and the FNR can be reduced by increasing epsilon, the authors should show such case in the article.

  We apologize for the error. We inadvertently repeated the same result. We have corrected the manuscript.

- The innovation in the proposed method for blocking relies on searching for nearest neighbours based either on shingles of the concatenated information contained in the records (1), or on embeddings of the information contained in the records (2). Several approximate nearest neighbour algorithms are made available in the package (nnd, hnsw, annoy, lsh, kd, klsh). What is the impact of the size of the shingles in approach (1)? How different are the blocking outputs with approach (1) or with approach (2)? Only approach (1) is illustrated in the vignettes. How missing values affect the construction of shingles? or the construction of embeddings?

  The paper is already quite long so this is why we did not decide to discuss impact of the number of shingles. We have added reference to Steorts et al. (2014) paper on LSH and shingles.

To provide answer to your question we have conducted a comparative analysis of the approaches using the `foreigners` dataset. The code below compares 2- and 3-character shingles, as well as embeddings generated with the `ragnar` package. This approach is versatile, as the package allows users to directly assign an embedding matrix to `x` and `y`. This action automatically sets `representation = "custom_matrix"` without requiring additional parameter specification (included in Supplementary Materials).

```
library(blocking)
library(data.table)
library(ragnar)

data("foreigners")
setDT(foreigners)

foreigners[, txt:=paste(fname, sname, surname, date, country, region)]
foreigners_emb <- embed_ollama(
  data.frame(text = foreigners$txt),
  base_url = "http://localhost:11434",
  model = "embeddinggemma",
  batch_size = 10L)

foreigners[, count:=.N, true_id]
foreigners[, row_id:=1:.N]
foreigners_sub <- foreigners[, head(.SD, 1), true_id]
foreigners_query <- foreigners[!row_id %in% foreigners_sub$row_id]

foreigners_matches <- merge(x = foreigners_sub[, .(x=1:.N, true_id)],
                            y = foreigners_query[, .(y = 1:.N, true_id)],
                            by = "true_id")
foreigners_matches[, block:=as.numeric(as.factor(true_id))]

foreigners_shingles_result <- blocking(x = foreigners_sub$txt,
                                       y = foreigners_query$txt,
                                       true_blocks = foreigners_matches[, .(x, y, block)],
                                       n_threads = 8,
                                       seed = 2026)

foreigners_shingles_result3 <- blocking(x = foreigners_sub$txt,
                                        y = foreigners_query$txt,
                                        true_blocks = foreigners_matches[, .(x, y, block)],
                                        n_threads = 8,
                                        seed = 2026,
                                        control_txt = controls_txt(n_shingles = 3L))

foreigners_sub_emb <- foreigners_emb$embedding[foreigners_sub$row_id, ]
foreigners_query_emb <- foreigners_emb$embedding[foreigners_query$row_id, ]

colnames(foreigners_sub_emb) <- colnames(foreigners_query_emb) <-
  paste0("v", 1:ncol(foreigners_query_emb))

foreigners_emb_result <- blocking(x = foreigners_sub_emb,
                                  y = foreigners_query_emb,
                                  true_blocks = foreigners_matches[, .(x, y, block)],
                                  n_threads = 8,
                                  seed = 2026)
```

The results indicate that increasing the shingle size to 3 leads to lower performance across all evaluation metrics. We have observed similar results while testing the

package with other datasets. In this specific case, the approach with embeddings yields poor results. However, with other datasets such as `census` and `cis` (also utilized in the package vignette, entitled *Blocking records for record linkage*), the embedding-based approach outperforms the methods utilizing shingles.

Here, we set `n_threads = 8`, which unfortunately does not allow for reproducing the exact same results due to an internal mechanism of the `rnndescent` package. However, the results obtained in different code runs should be very close to those presented here.

```
########################
### shingles = 2
########################
> foreigners_shingles_result
==========================================================
Blocking based on the nnd method.
Number of blocks: 6431.
Number of columns used for blocking: 1232.
Reduction ratio: 0.9999.
==========================================================
Distribution of the size of the blocks:
   2    3    4    5    6    7
3857 1605  946   21    1    1
==========================================================
Evaluation metrics (standard, in %):
     recall   precision         fpr         fnr    accuracy specificity    f1_score
    96.9956     79.4200      0.0037      3.0044     99.9959     99.9963     87.3323

########################
### shingles = 3
########################

> foreigners_shingles_result3
==========================================================
Blocking based on the nnd method.
Number of blocks: 6509.
Number of columns used for blocking: 9020.
Reduction ratio: 0.9999.
==========================================================
Distribution of the size of the blocks:
   2    3    4    5    6
3989 1591  895   26    8
==========================================================
Evaluation metrics (standard, in %):
     recall   precision         fpr         fnr    accuracy specificity    f1_score
    96.2415     76.8200      0.0042      3.7585     99.9953     99.9958     85.4410

########################
### embeddings
########################


> foreigners_emb_result
==========================================================
Blocking based on the nnd method.
Number of blocks: 7050.
Number of columns used for blocking: 768.
Reduction ratio: 0.9999.
==========================================================
Distribution of the size of the blocks:
```

```
   2    3    4    5    6    7    8    9   10   11   13
5063 1359  458   94   36   18    9    4    6    2    1
=========================================================
Evaluation metrics (standard, in %):
     recall   precision         fpr         fnr    accuracy specificity    f1_score
    79.1361     45.2500      0.0098     20.8639     99.9881     99.9902     57.5773
```

Regarding the missing values: since we concatenate all key variables into a single string before generating shingles or embeddings, missing values are treated as empty strings.

- No comment is made on Figure 1. This figure though is a nice illustration of the entity resolution problem and of the fact that blocking may eliminate some pairs of records which are matches in reality due to large distances between the records (the second red mode is not distinguishable from the blue one in real applications where true labels are unknown). Moreover, the situation could be even more complex; in hard record linkage or deduplication tasks, where a lot of FP are expected (lots of records contain similar information due to low discriminative power of the identifying information though they are non-matches), both matches and non-matches distributions might be bimodal.

  We have followed this suggestion and added a detailed description of Figure 1 in the text, explaining the implications of the presented distributions for the blocking process.

- The evaluation metrics for entity resolution are usually expressed as values between 0 and 1. The way it is presented now in Table 2 might confuse the reader. For instance, the precision of 5.17 might seem different from the precision of 4.76 though they actually correspond both to precisions of 0.05 (5%). If presenting the metrics as percentages seems more relevant, please specify that these are expressed in percentage.

  We have decided to present the evaluation metrics as percentages. We have updated the text to explicitly specify this.

- The comments associated to Table 2 are confusing. All the methods provide high recall, low precision, high specificity, as expected. Indeed the traditional tree-based ANN algorithms seem flexible and promising for entity resolutions. The phrasing though puts into comparisons all the different methods which is not the meaning of the package I think. Do the confusion matrices differ between the methods? It might be relevant to show it if so.

  We have rephrased the discussion around Table 2 to make it clear that the results constitute a straightforward empirical comparison of the available methods, rather than a recommendation of one over another. The package is intended as a toolkit and the comparison is provided solely to illustrate the behaviour of the methods on the example dataset.

  Regarding the confusion matrices, we have moved the code generating the table to the Supplementary materials. As suggested by the Reviewer, the updated code now outputs confusion matrices for all compared methods.

- What is the 'computational cost' of the implemented methods on the running example? It would be relevant to mention time and memory use (and characteristics of the computer).

All methods included in the example execute in less than 0.6 seconds (MacBook Pro, Apple M4 Pro 14-core CPU, 48 GB RAM). Given the small size of the `RLdata500` dataset, we consider these costs negligible. Therefore, we have decided not to include detailed performance metrics for this example in the manuscript.

- By using shingles or embeddings, the meaning of numerical information contained in the records will likely be lost. Is it possible to apply different strategies to the different identifying variables contained in the records? For instance, the search of nearest neighbours could be employed on

  - shingles or embeddings of the character variables,
  - binary comparisons or distances between the numerical variables,
  - other more complex blocking rules depending on the nature of the variables (like in SPLink or dedupe: similar 3 first characters, close integers, common four gram, Levenshtein distance between variables...) In that regard, is it possible for the users to provide their own method, to compare it with the implemented methods in the explorative phase of entity resolution?

  We agree that combining different blocking strategies for different variable types (e.g., shingles or embeddings for character variables, distances for numerical variables) would be a valuable extension. However, this functionality has not yet been implemented in the current version of the package. The general philosophy behind the package is to provide a straightforward, easy-to-use tool that requires minimal configuration from the user. We consider this a promising direction for future development and will note it as such in the paper.

  If Reviewer is interested we have done similar study with the `BlockingPy` and `splink` and it seems that it improved linking (see `https://blockingpy.readthedocs.io/en/latest/examples/splink_integration.html`).

  Our recommendation would be to use results of the 'blocking()' function as additional blocking variable in a standard blocking procedure (as in the `BlockingPy` and `splink` example).

- In the introduction, the authors mention that the package integrates well with the reclin2 R package for record linkage. However, the package is not restricted to be used with that package and it seems broadly applicable to any record linkage or deduplication task. (I come back to this topic in a comment on the code).

  We agree with this observation. We have updated the Introduction to state that the package is suitable for any record linkage or deduplication workflow.

- The authors implemented the 'est_block_error' function with the FPR and FNR estimators proposed by Dasylva and Goussanou. I think this nice addition worths an illustration in the article.

  We appreciate the positive feedback on this feature. However, since it is auxiliary to the main package functionality, we have decided not to include a specific illustration in the manuscript. Instead, we have improved the function's help file, which now clearly describes its usage.

- The authors show in the vignettes how to use the blocking package with reclin2, fastLink and RecordLinkage. Would it be possible to present a broadly applicable example where blocking is performed and a record linkage algorithm or an entity resolution algorithm is then employed on the data set(s) reduced thanks to blocking. (Maybe simply a for loop on the blocks output by the package on which any record linkage or record deduplication package can be applied?)

  We have updated the example in Section 3.1 to include the Expectation-Maximization algorithm by Fellegi and Sunter. This demonstrates a complete entity resolution workflow based on the generated blocks.

- The authors show in the vignette results obtained using 2 characters shingles to search for nearest neighbours. Is it possible to illustrate the use of embeddings representing the input character vectors?

  As detailed in our response to one of the previous comments, we have prepared a code example comparing shingle- and embedding-based approaches. We have also added this example to the Supplementary materials to serve as a practical illustration for users interested in using embeddings.

- The function 'est_block_error' is not documented in the vignettes and does return an error when applied on the data provided in the package for the record linkage task, specifically: Error in if (abs(log_lik_new - log_lik_old) <= tol) { : missing value where TRUE/FALSE needed In addition: Warning message: In dpois(n - 1, y) : NaNs produced Moreover, some guidance on the parameter G would be useful, I assume it would take value 2 or 3 for matches, non-matches and potentially uncertain cases.

  We have added a comprehensive example in the package documentation, showing how to use the `est_block_error` function directly with the output of `blocking`. While we were unable to reproduce the specific error reported by the Reviewer, we trust that the updated documentation and examples clarify the correct usage.

  Regarding the parameter `G`, following a discussion with Abel Dasylva, we have implemented an automatic selection of `G` based on the Akaike Information Criterion in the revised package version.

4 Reviewer 3

- Thank you for submitting the paper and producing the package. Overall, a good paper, but only useful for experienced linkers. It is strange to see Soundex explained, but the technical details are not explained. The paper should contain more technical details (I mean, text and equations) and more comments in the listings. I miss a clear statement of if and how PPRL could be used with the package. Most importantly, what are the limits of files to be used? The size of your input files is in many applications in official statistics the size of the resulting blocks. The maximum size of the input files should cover populations in Europe, so we would see files with 100 million records each. In practice, most RL programs are useless with files of this size; therefore, we block. Can you block two 100 million record files with the package?

  In principle it is possible to do so with the `nndescent` package but this would require a lot of RAM. In principle, it possible to save an index that RcppHNSW (see) or RcppAnnoy (see `https://cran.r-project.org/web/packages/RcppAnnoy/refman/RcppAnnoy.html#AnnoyIndex`) on a disk and then query the index with a new dataset. However, this approach would require a custom script that saves chunks of the index on a disk.

  We have experience with using blocking package with the `nndescent` package on blocking population register in Poland (limited to foreign-nationals, about 4 mln records) with insurance register (same subset, about 1 mln) and it worked smoothly. The main reason was that we were using shingles (so sparse matrix) instead of a dense matrix.

  Possible way around, that is not yet implemented in our package, is to use duckdb and HNSW / Faiss module (see `https://duckdb.org/community_extensions/extensions/faiss`) but we have not tried this approach (there is a note that "The index itself is not buffer managed and must be able to fit into RAM memory.").

  Regarding PPRL we did not have a solution yet, we have not thought about that when desiging blocking. However, this sounds as an interesting idea to follow. We added a paragraph on PPRL in the summary: *The package does not currently support Privacy-Preserving Record Linkage; however, the flexibility of the 'representation' argument (which accepts user-supplied matrices) provides a natural entry point for integrating encodings such as Bloom filters. Developing and evaluating dedicated PPRL workflows is a direction for future work.*

- Page 1, Line 3 "without common identifiers" should be "without common unique identifiers" "This drives the need to link records without identifiers" -> "This drives the need to link records without unique identifiers..."

  We have revised the manuscript accordingly.

- Wright 1960 as a source for Soundex. Please use either the original citation, or Knuth or Christen as standard references.

  Corrected, we used Knuth, D. E. (1973). The Art of Computer Programming, Vol. 3: Sorting and Searching. Addison-Wesley. (pp. 391–392)

- If "one may consider approaches proposed by Dasylva et al.", why do you not consider it here? Either give a reasoning or drop the sentence.

Now it is as follows: Alternatively, one may consider approaches proposed by Dasylva and Goussanou (2021) and Dasylva and Goussanou (2022), who demonstrated how to estimate FPR and FNR without access to an audit sample.

- Page 2 If you explain basics such as Soundex, you should explain shingles.

  We provided an example in the text "... (n-character sequences, e.g. `"home"` will be `c("ho", "om", "me")`)"

- In the same sentence on page 2 ("using tokenizers (Mullen et al., 2018) and text2vec (Selivanov et al., 2023) packages, or provide a matrix of vectors (e.g., embeddings via ragnar, Kalinowski and Falbel (2025)) representing the input character vectors.") This is too dense. If I understand it correctly, you use either tokenizers or text2vec, not and. And then you have to explain that they do. The same is true for embeddings. Give a sentence to explain what an embedding is and another sentence about what ragnar is.

  In the software we use both: first we use tokenizers to create shingles and we follow with text2vec, see: `https://github.com/ncn-foreigners/blocking/blob/194d17f3cc2740939c7ded06d17475a1528d5f78/R/blocking.R#L220`

- And here you please also state if your library is suitable for PPRL, and for what kinds of PPRL.

  We thank the Reviewer for this question. We must admit that PPRL was not considered during the development of the package or the writing of this paper. Having looked into this after receiving the review, we note that if Bloom filter encodings (as produced by the `CreateBF()` function in the `PPRL` package) are expanded into binary columns (one column per bit position), then ANN-based blocking using Hamming distance or Jaccard similarity on binary vectors would be meaningful and compatible with the package. We have added a note to the paper acknowledging this potential use case and identifying full PPRL compatibility as a direction for future work.

- Explain what ANN algorithms are.

  We have added the following explanation: "ANN algorithms are methods designed to efficiently find the closest matches to a query point in a large dataset by trading off search accuracy for computational speed. Unlike exact nearest neighbour search, ANN algorithms do not guarantee finding the true nearest neighbours but are able to identify close approximations substantially faster (cf. Indyk and Motwani (1998), Wang et al. (2021)), making them naturally suited to noisy or incomplete data."

- In a text for a statistical audience, "significantly" should only be used in a technical way. You mean "substantially", I think.

  We have revised the manuscript accordingly.

- "as described in one of the package vignettes": In which? Should have a name or number.

  We have followed this recommendation and updated the text.

- page 3 1.3 Outline of article -> Outline of the article

We have revised the manuscript accordingly.

- "how results can be assessed." I'm not sure, what you mean by that part of the sentence. Do you mean "evaluated"?

  We have followed this recommendation and updated the text.

- "x, y – reference vectors" : What are "reference vectors"? You mean the vector of keys used for linkage?

  Yes, we have revised the manuscript accordingly.

- " indicates that deduplication is applied;" -> indicates a deduplication task

  We have followed this recommendation and updated the text.

- Page 4 You should give a reference for the reduction ratio. And for the other metrics. Please consider the critique of Christen and Hand on F1. Maybe implementing their suggestion is an option.

  Added with a comment that we will include F1 proposal in the future.

- Page 5 last line (result_reclin) should be on page 6

  We have corrected the formatting to keep this code block on the same page.

- Page 7 "Below, we present an example using the reclin2 package with a simple score." Explain the "simple score" part of the sentence. The example does not make it clearer. What exactly is the last column in the output? Do we have an equation?

  We have revised this example. We have replaced the simple score (which was a sum of Jaro-Winkler similarities for `fname`, `surname`, and `date`) with the Expectation-Maximization algorithm by Fellegi and Sunter.

- page 9: Comment in more detail what the code does:

```
df_block_melted <- melt(res_dedup$result, id.vars = c("block", "dist"))
df_block_melted_rec_block <- unique(df_block_melted[, .(rec_id=value,  block)])
RLdata500[df_block_melted_rec_block, on = "rec_id", block_id :=  i.block]
RLdata500[, .(uniq_blocks = uniqueN(block_id)), .(ent_id)][, .N, uniq_blocks]
```

  A few comments might make reading easier; otherwise, you could simply publish listings.

  We have followed this recommendation and added comments within the code block.

- Page 10 is a listing without comments. This is not the style of an article. You should explain the job, not print a listing.

  We agree with the Reviewer. Since the example is focused on the results rather than the code itself, we have decided to move the listing to the Supplementary materials to improve readability.

- Page 11 Precision and recall are usually scaled to 0-1. In the table, we have larger values.

  We have decided to present the evaluation metrics as percentages. We have updated the text to explicitly specify this.